

Lecture 6: The Newton Convergence Theorem

Math 6610

C. Mou

Department of Mathematics and Statistics
Utah State University

Fall 2026

Today's Goals

- State the Newton–Kantorovich-type convergence theorem precisely

Today's Goals

- State the Newton–Kantorovich-type convergence theorem precisely
- Understand the roles of α (first step), γ (nonlinearity), and $h = \alpha\gamma < \frac{1}{2}$

Today's Goals

- State the Newton–Kantorovich-type convergence theorem precisely
- Understand the roles of α (first step), γ (nonlinearity), and $h = \alpha\gamma < \frac{1}{2}$
- Follow the induction: iterates stay in the ball, Jacobians stay nonsingular

Today's Goals

- State the Newton–Kantorovich-type convergence theorem precisely
- Understand the roles of α (first step), γ (nonlinearity), and $h = \alpha\gamma < \frac{1}{2}$
- Follow the induction: iterates stay in the ball, Jacobians stay nonsingular
- Master the two important tools: the Banach lemma and the integral form of the mean value theorem

The Newton Convergence Theorem: Assumptions

Theorem. Given:

- $D \subset \mathbb{R}^n$ convex, $f : D \rightarrow \mathbb{R}^n$ differentiable on D
- $x^0 \in D$ and $Df(x^0)$ is nonsingular (invertible)
- $\|Df(x^0)^{-1}(Df(y) - Df(x))\| \leq \gamma \|y - x\|$ for all $x, y \in D$, with $\gamma > 0$
(bigger $\gamma \Rightarrow$ more nonlinear)
- $\|Df(x^0)^{-1}f(x^0)\| \leq \alpha$ for some $\alpha > 0$
- $h := \alpha\gamma < \frac{1}{2}$
- $\overline{S_r(x^0)} \subset D$, $r = \frac{1 - \sqrt{1 - 2h}}{\gamma} > 0$ (balls: next slide)

Claims: two slides ahead.

Remark. Df is automatically continuous on D : for $x, y \in D$,

$$\|Df(y) - Df(x)\| = \|Df(x^0) Df(x^0)^{-1} (Df(y) - Df(x))\| \leq \|Df(x^0)\| \gamma \|y - x\|.$$

The integral in the proof needs this.

The Ball Around x^0

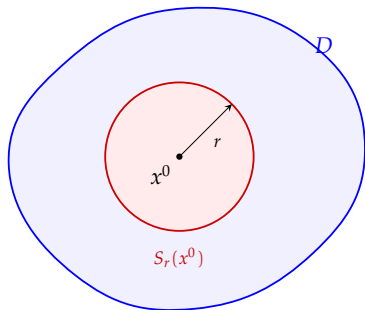
Definition (*Open and closed ball*).

$$S_r(x^0) := \{x \in \mathbb{R}^n \mid \|x - x^0\| < r\}, \quad \overline{S_r(x^0)} := \{x \in \mathbb{R}^n \mid \|x - x^0\| \leq r\}.$$

Both balls are convex: for x, y in the ball and $0 \leq t \leq 1$,

$$\begin{aligned} & \|x + t(y - x) - x^0\| \\ &= \|(1 - t)(x - x^0) + t(y - x^0)\| \\ &\leq (1 - t) \|x - x^0\| + t \|y - x^0\| \\ &< r \quad (\text{resp. } \leq r). \end{aligned}$$

Assumption of the theorem: $\overline{S_r(x^0)} \subset D$. So every segment between two points of the ball lies in D .



The Newton Convergence Theorem: Claims

Theorem (*continued*). Claims:

- 1 All Newton iterates x^k are defined and satisfy $x^k \in S_r(x^0)$, $Df(x^k)$ is nonsingular
- 2 The iterates x^k converge quadratically to a zero of f , $x^* := \lim_{k \rightarrow \infty} x^k$,
 $f(x^*) = 0$, $x^* \in \overline{S_r(x^0)}$
- 3 x^* is the only zero of f in $D \cap S_R(x^0)$, $R := \frac{1 + \sqrt{1 - 2h}}{\gamma}$ ($R > r$)

Remark (*The first step*). The Newton equation $Df(x^0) \Delta x^0 = -f(x^0)$ gives

$$\|x^1 - x^0\| = \|-Df(x^0)^{-1}f(x^0)\| \leq \alpha = \frac{h}{\gamma}.$$

So α is a bound on the length of the first Newton step.

Two Tools (1/2): The Banach Lemma

Used for: showing that $Df(x^k)$ stays nonsingular when x^k is close to x^{k-1} – a small perturbation of a nonsingular matrix is nonsingular.

Lemma (Banach). If $A \in \mathbb{R}^{n \times n}$ and $\|A\| < 1$, then $I - A$ is nonsingular and $\|(I - A)^{-1}\| \leq \frac{1}{1 - \|A\|}$.

Proof. 1. Nonsingular. If $(I - A)v = 0$ then $v = Av$, so

$$\|v\| = \|Av\| \leq \|A\| \|v\| \implies (1 - \|A\|) \|v\| \leq 0 \implies v = 0.$$

A square matrix $I - A$ with $(I - A)v = 0$ only for $v = 0$ is nonsingular.

2. Bound. Let $B := (I - A)^{-1}$. From $(I - A)B = I$ we get $B = I + AB$, so

$$\|B\| \leq \|I\| + \|A\| \|B\| = 1 + \|A\| \|B\| \implies \|B\| \leq \frac{1}{1 - \|A\|}.$$



Remark (Neumann series). $(I - A)^{-1} = I + A + A^2 + A^3 + \dots$. The series converges because $\|A^j\| \leq \|A\|^j$ and $\sum_{j \geq 0} \|A\|^j = \frac{1}{1 - \|A\|}$; multiplying it by $I - A$ telescopes to I . This gives the same bound.

Two Tools (2/2): The Integral Identity

Used for: comparing f at two iterates; it turns $f(x^k)$ into an integral of differences of Jacobians, which the Lipschitz condition bounds.

Lemma (Mean value theorem, integral form). Let Df be continuous on D , $x, y \in D$, $\Delta := y - x$, and let the segment $x + t\Delta$, $0 \leq t \leq 1$, lie in D . Then

$$f(y) - f(x) = \int_0^1 Df(x + t\Delta) \Delta dt \quad (\text{integral taken entry by entry}).$$

Moreover, for continuous $v : [0, 1] \rightarrow \mathbb{R}^n$: $\left\| \int_0^1 v(t) dt \right\| \leq \int_0^1 \|v(t)\| dt$.

Proof. Set $\varphi(t) := f(x + t\Delta)$ for $0 \leq t \leq 1$. By the chain rule, $\varphi'(t) = Df(x + t\Delta) \Delta$, which is continuous in t . The fundamental theorem of calculus, applied to each of the n entries, gives

$$f(y) - f(x) = \varphi(1) - \varphi(0) = \int_0^1 \varphi'(t) dt.$$

For the inequality: every Riemann sum satisfies $\|\sum_i v(t_i) \Delta t_i\| \leq \sum_i \|v(t_i)\| \Delta t_i$ by the triangle inequality; let the mesh go to 0 (the norm is continuous). ■

Proof of Parts 1 and 2: The Idea (1/2)

Remark (Intuitions). Newton's method can fail in three ways: an iterate leaves D (f undefined); some $Df(x^k)$ is singular (no step); the steps do not shrink (no convergence). Each step exists only if the previous one succeeded, so the guarantees must be established one step at a time: *induction on k* .

At x^0 we control two numbers: α (length of the first step) and γ (how fast Df changes, seen through $Df(x^0)^{-1}$). The proof tracks the same two numbers at every iterate:

- γ_k : the Lipschitz constant seen from x^k , through $Df(x^k)^{-1}$.
- h_k/γ_k : a bound on the step $\|x^{k+1} - x^k\|$. It *shrinks fast*: each step is quadratic in the previous one.
- $h_k = (\text{step bound}) \times (\text{nonlinearity})$: the product $h = \alpha\gamma$ recomputed at x^k . $h_k < \frac{1}{2}$ means the tangent model at x^k is still trustworthy.

Proof of Parts 1 and 2: The Idea (2/2)

Remark (Why a ball, and why this radius). The iterates never leave D if they never leave a ball around x^0 that lies in D . The distance from x^0 is at most the total length of the steps taken:

$$\|x^k - x^0\| \leq \sum_{j=0}^{k-1} \|x^{j+1} - x^j\| \leq \sum_{j=0}^{k-1} \frac{h_j}{\gamma_j}.$$

So the radius must dominate the total travel. The Lemma below shows

$$\sum_{j=0}^{k-1} \frac{h_j}{\gamma_j} = \frac{1}{\gamma} - \frac{1}{\gamma_k} \nearrow \frac{1 - \sqrt{1 - 2h}}{\gamma} = r \quad (k \rightarrow \infty) :$$

the total-travel bound increases with k , stays strictly below r for every k , and tends to r . That is where the formula for r comes from.

Remark (Plan). (1) Define h_k, γ_k and prove their properties: pure algebra. (2) Induction on k : x^k in the ball, $Df(x^k)$ nonsingular with constant γ_k , step $\leq h_k/\gamma_k$. (3) Next lecture: Cauchy $\Rightarrow$ limit x^* ; $f(x^*) = 0$; rate; uniqueness.

Proof of Parts 1 and 2: The Majorant Sequences

Proof of parts 1 and 2. The vector iteration is controlled by two sequences of numbers.

Definition (*Majorant sequences*).

$$h_k := \begin{cases} h & k = 0 \\ \frac{1}{2} \frac{h_{k-1}^2}{(1 - h_{k-1})^2} & k > 0 \end{cases} \quad \gamma_k := \begin{cases} \gamma & k = 0 \\ \frac{\gamma_{k-1}}{1 - h_{k-1}} & k > 0 \end{cases}$$

Remark (*Reading the symbols*). γ_k = nonlinearity seen from x^k ; h_k/γ_k = bound on the step $\|x^{k+1} - x^k\|$; h_k = their product, i.e. the quantity $h = \alpha\gamma$ of the theorem recomputed at x^k .

Where the Recursions Come From

Remark (*The definitions are read off from the proof*). Suppose at stage $k - 1$ we know the constant γ_{k-1} and the step bound $\|x^k - x^{k-1}\| \leq h_{k-1}/\gamma_{k-1}$. The two estimates of the induction step (slides ahead) will give

$$\gamma_k = \frac{\gamma_{k-1}}{1 - h_{k-1}} \quad (\text{Banach lemma: the inverse grows by at most this factor}),$$

$$\|x^{k+1} - x^k\| \leq \frac{\gamma_k}{2} \|x^k - x^{k-1}\|^2 \leq \frac{\gamma_k}{2} \left(\frac{h_{k-1}}{\gamma_{k-1}} \right)^2 \quad (\text{quadratic estimate}).$$

Define h_k so that the last bound reads h_k/γ_k :

$$h_k := \gamma_k \cdot \frac{\gamma_k}{2} \left(\frac{h_{k-1}}{\gamma_{k-1}} \right)^2 = \frac{1}{2} \left(\frac{\gamma_k}{\gamma_{k-1}} h_{k-1} \right)^2 = \frac{1}{2} \frac{h_{k-1}^2}{(1 - h_{k-1})^2}.$$

So the recursions are not guessed: they are exactly what makes the induction close from one stage to the next.

Properties of the Majorant Sequences

Lemma (*Properties of h_k, γ_k*). For all $k = 0, 1, \dots$:

- (i) $0 < h_{k+1} < h_k < \frac{1}{2}$ and $\lim_{j \rightarrow \infty} h_j = 0$
- (ii) $\gamma_k < \gamma_{k+1} < \frac{\gamma}{\sqrt{1-2h}} = \lim_{j \rightarrow \infty} \gamma_j$
- (iii) $\frac{h_k}{\gamma_k} \leq \frac{1}{2^k} \alpha$
- (iv) $\frac{h_0}{\gamma_0} + \frac{h_1}{\gamma_1} + \dots + \frac{h_{k-1}}{\gamma_{k-1}} = \frac{1}{\gamma} - \frac{1}{\gamma_k} < r = \frac{1 - \sqrt{1-2h}}{\gamma}$

Remark (*What each property is for*). (i) the tangent model stays trustworthy at every iterate, and ever more so. (ii) the Lipschitz constants stay bounded, so the inverses $Df(x^k)^{-1}$ never blow up; the limit $\gamma / \sqrt{1-2h}$ is what produces the formula for r . (iii) the steps decay at least geometrically: this will make (x^k) a Cauchy sequence. (iv) the total travel stays below r : the iterates stay in the ball.

Proof of the Lemma (1/2): (i) and (ii)

Proof. (i) Induction. Let $0 < h_{k-1} < \frac{1}{2}$. Then $0 < \frac{h_{k-1}}{1-h_{k-1}} < 1$ and $(1-h_{k-1})^2 > \frac{1}{4}$, so

$$0 < h_k = \frac{1}{2} \left(\frac{h_{k-1}}{1-h_{k-1}} \right)^2 < \frac{1}{2}, \quad h_k < 2h_{k-1}^2 < h_{k-1}.$$

Also $2h_k < (2h_{k-1})^2$, hence $2h_k < (2h)^{2^k} \rightarrow 0$, because $2h < 1$.

(ii) $\gamma_k = \gamma_{k-1}/(1-h_{k-1}) > \gamma_{k-1}$ since $0 < h_{k-1} < 1$. For the bound, compute

$$1 - 2h_k = 1 - \frac{h_{k-1}^2}{(1-h_{k-1})^2} = \frac{1-2h_{k-1}}{(1-h_{k-1})^2} \implies \sqrt{1-2h_k} = \frac{\sqrt{1-2h_{k-1}}}{1-h_{k-1}}.$$

Dividing by $\gamma_k = \gamma_{k-1}/(1-h_{k-1})$, the factors $1-h_{k-1}$ cancel:

$$\frac{\sqrt{1-2h_k}}{\gamma_k} = \frac{\sqrt{1-2h_{k-1}}}{\gamma_{k-1}} = \dots = \frac{\sqrt{1-2h}}{\gamma}, \quad \text{i.e.} \quad \gamma_k = \gamma \frac{\sqrt{1-2h_k}}{\sqrt{1-2h}}.$$

Since $0 < h_k \rightarrow 0$: $\gamma_k < \gamma/\sqrt{1-2h}$ and $\gamma_k \rightarrow \gamma/\sqrt{1-2h}$. (continued)

Proof of the Lemma (2/2): (iii) and (iv)

Proof (continued). (iii) By the definitions and $\frac{h_{k-1}}{1-h_{k-1}} < 1$,

$$\frac{h_k}{\gamma_k} = \frac{1}{2} \frac{h_{k-1}^2}{(1-h_{k-1})^2} \cdot \frac{1-h_{k-1}}{\gamma_{k-1}} = \frac{h_{k-1}}{\gamma_{k-1}} \cdot \frac{1}{2} \frac{h_{k-1}}{1-h_{k-1}} < \frac{1}{2} \frac{h_{k-1}}{\gamma_{k-1}}.$$

With $h_0/\gamma_0 = h/\gamma = \alpha$, induction gives $h_k/\gamma_k \leq \alpha/2^k$.

(iv) From $1/\gamma_{j+1} = (1-h_j)/\gamma_j$,

$$\frac{h_j}{\gamma_j} = \frac{1}{\gamma_j} - \frac{1}{\gamma_{j+1}} \quad \Longrightarrow \quad \sum_{j=0}^{k-1} \frac{h_j}{\gamma_j} = \frac{1}{\gamma_0} - \frac{1}{\gamma_k} = \frac{1}{\gamma} - \frac{1}{\gamma_k} \quad (\text{the sum telescopes}).$$

By (ii), $1/\gamma_k > \sqrt{1-2h}/\gamma$, so

$$\frac{1}{\gamma} - \frac{1}{\gamma_k} < \frac{1 - \sqrt{1-2h}}{\gamma} = r.$$

The Scalar Picture: Newton on a Parabola (1/2)

Remark (One dimension). Take the simplest function with the theorem's data – first Newton step α from $t = 0$, and derivative with Lipschitz constant γ :

$$p(t) := \frac{\gamma}{2} t^2 - t + \alpha, \quad -\frac{p(0)}{p'(0)} = \alpha, \quad |p'(s) - p'(t)| = \gamma |s - t|.$$

Its zeros are exactly r and R :

$$p(t) = 0 \iff t = \frac{1 \pm \sqrt{1 - 2\alpha\gamma}}{\gamma} = \frac{1 \pm \sqrt{1 - 2h}}{\gamma} \in \{r, R\},$$

and they are real and distinct exactly when $1 - 2h > 0$, i.e. $h < \frac{1}{2}$. So the condition $h < \frac{1}{2}$ says: *the worst-case parabola still has a zero*, and r, R are its two zeros.

The Scalar Picture: Newton on a Parabola (2/2)

Remark (*The majorant sequences are Newton's method on p*). Run Newton's method on p from $t_0 := 0$. Then $t_1 = \alpha$, and

$$t_k = \frac{1}{\gamma} - \frac{1}{\gamma_k}, \quad t_{k+1} - t_k = \frac{h_k}{\gamma_k}, \quad t_k \nearrow r.$$

Check: $p'(t_k) = \gamma t_k - 1 = -\gamma/\gamma_k$, and with $\gamma_k = \gamma \sqrt{1-2h_k}/\sqrt{1-2h}$ (Lemma (ii)) one computes $p(t_k) = \gamma h_k/\gamma_k^2$; so the Newton step is $-p(t_k)/p'(t_k) = h_k/\gamma_k$, and $t_k = \sum_{j < k} h_j/\gamma_j$ by Lemma (iv).

The induction below shows

$$\|x^{k+1} - x^k\| \leq \frac{h_k}{\gamma_k} = t_{k+1} - t_k:$$

each step of the vector iteration is no longer than the corresponding step of Newton's method on the parabola. Hence the total travel is at most $\lim t_k = r$, the iterates stay in $S_r(x^0)$, and they converge at least as fast as Newton on p – quadratically, since r is a simple zero ($p'(r) = -\sqrt{1-2h} \neq 0$). The larger zero R marks the region where Part 3 guarantees uniqueness.

Proof of Parts 1 and 2: The Induction Claim

Claim: For $k = 0, 1, \dots$,

(a) $x^k \in S_r(x^0)$ (so $x^k \in D$, and $f(x^k)$, $Df(x^k)$ are defined)

(b) $Df(x^k)$ is nonsingular (so x^{k+1} is defined)

(c) $\|Df(x^k)^{-1}(Df(y) - Df(x))\| \leq \gamma_k \|y - x\|$ for all $x, y \in D$

(d) $\|x^{k+1} - x^k\| \leq \frac{h_k}{\gamma_k} \leq \frac{1}{2^k} \alpha$

(a) and (b) for all k are Part 1 of the theorem; (d) is what Part 2 is built on.

Why all four together: (a) and (b) make the next step exist, (c) is the tool that bounds it, (d) is the bound – and (d) feeds back into (a) and (c) at the next stage.

Proof by induction on k .

$k = 0$: (a) $\|x^0 - x^0\| = 0 < r$; (b) and (c) are assumptions of the theorem ($\gamma_0 = \gamma$); (d) is the first-step remark, $\|x^1 - x^0\| \leq \alpha = h_0/\gamma_0$.

$k \geq 1$: assume the Claim holds for $0, 1, \dots, k-1$. We prove (a)–(d) for k on the next five slides.

Induction Step (1/5): x^k Stays in the Ball

Why: the distance from x^0 is at most the total length of the steps, and Lemma (iv) keeps that total below r .

(a) Triangle inequality, then (d) for $0, \dots, k-1$, then Lemma (iv):

$$\|x^k - x^0\| \leq \underbrace{\|x^k - x^{k-1}\|}_{\leq \frac{h_{k-1}}{\gamma_{k-1}}} + \underbrace{\|x^{k-1} - x^{k-2}\|}_{\leq \frac{h_{k-2}}{\gamma_{k-2}}} + \dots + \underbrace{\|x^1 - x^0\|}_{\leq \frac{h_0}{\gamma_0}} \leq \frac{h_0}{\gamma_0} + \frac{h_1}{\gamma_1} + \dots + \frac{h_{k-1}}{\gamma_{k-1}} <$$

Thus $x^k \in S_r(x^0) \subset D$: the function and its Jacobian are defined at x^k , and every segment from x^{k-1} to x^k lies in D .

Induction Step (2/5): $Df(x^k)$ as a Perturbation

Why: we know $Df(x^{k-1})$ is nonsingular. Since x^k is close to x^{k-1} and Df is Lipschitz, $Df(x^k)$ differs from $Df(x^{k-1})$ by a small *relative* amount A – measured, as always, through the inverse $Df(x^{k-1})^{-1}$.

Factor out the known nonsingular matrix:

$$Df(x^k) = \underbrace{Df(x^{k-1})}_{\text{nonsingular}} \left(I - \underbrace{Df(x^{k-1})^{-1} (Df(x^{k-1}) - Df(x^k))}_{:=A} \right) \quad (1)$$

(Multiply out: $Df(x^{k-1}) - (Df(x^{k-1}) - Df(x^k)) = Df(x^k)$.) By (c) and (d) for $k-1$, with $x = x^{k-1}$, $y = x^k$ (both in D):

$$\|A\| = \|Df(x^{k-1})^{-1} (Df(x^k) - Df(x^{k-1}))\| \leq \gamma_{k-1} \underbrace{\|x^k - x^{k-1}\|}_{\leq \frac{h_{k-1}}{\gamma_{k-1}}} \leq h_{k-1} < \frac{1}{2} < 1.$$

The relative perturbation is less than 1: this is what the Banach lemma needs.

Induction Step (3/5): Nonsingularity and γ_k

Why: a matrix within relative distance < 1 of a nonsingular one is nonsingular; its inverse is larger by a factor at most $1/(1 - \|A\|)$, and the Lipschitz constant seen from x^k grows by that same factor.

(b) Banach lemma (Tool 1): $\|A\| \leq h_{k-1} < 1$, so $I - A$ is nonsingular with $\|(I - A)^{-1}\| \leq \frac{1}{1 - h_{k-1}}$. By (1), $Df(x^k)$ is a product of two nonsingular matrices, hence nonsingular, and $Df(x^k)^{-1} = (I - A)^{-1} Df(x^{k-1})^{-1}$.
(c) For $x, y \in D$, by this formula, $\|MN\| \leq \|M\| \|N\|$, and (c) for $k - 1$:

$$\begin{aligned}\|Df(x^k)^{-1}(Df(y) - Df(x))\| &= \|(I - A)^{-1} Df(x^{k-1})^{-1}(Df(y) - Df(x))\| \\ &\leq \|(I - A)^{-1}\| \|Df(x^{k-1})^{-1}(Df(y) - Df(x))\| \\ &\leq \frac{1}{1 - h_{k-1}} \gamma_{k-1} \|y - x\| = \gamma_k \|y - x\|.\end{aligned}$$

Induction Step (4/5): Rewriting $x^{k+1} - x^k$

Why: the new step is $-Df(x^k)^{-1}f(x^k)$, so we need $f(x^k)$ to be small. It is: the Newton step from x^{k-1} made the *tangent model* vanish at x^k , so $f(x^k)$ is exactly the error of that tangent model – and Taylor tells us such an error is quadratic in the step.

Integral identity (Tool 2) with $x = x^{k-1}$, $y = x^k$ (the segment lies in the ball $\subset D$):

$$f(x^k) - f(x^{k-1}) = \int_0^1 Df(x^{k-1} + t(x^k - x^{k-1}))(x^k - x^{k-1}) dt.$$

The Newton equation for step $k-1$ says $f(x^{k-1}) + Df(x^{k-1})(x^k - x^{k-1}) = 0$, so

$$\begin{aligned} x^{k+1} - x^k &= -Df(x^k)^{-1} \underbrace{\left(f(x^k) - f(x^{k-1}) - Df(x^{k-1})(x^k - x^{k-1}) \right)}_{=0} \\ &= -Df(x^k)^{-1} \int_0^1 \left(Df(x^{k-1} + t(x^k - x^{k-1})) - Df(x^{k-1}) \right) (x^k - x^{k-1}) dt. \end{aligned}$$

Induction Step (5/5): The Step Bound

Why quadratic: the integrand is a *difference of Jacobians* times the old step. The Lipschitz condition makes the difference small (factor $\gamma_k t \|x^k - x^{k-1}\|$), and the old step is small (another factor $\|x^k - x^{k-1}\|$).

(d) Move $Df(x^k)^{-1}$ inside, take norms, use (c) for k with $y = x^{k-1} + t(x^k - x^{k-1})$, $x = x^{k-1}$, so $y - x = t(x^k - x^{k-1})$:

$$\begin{aligned}\|x^{k+1} - x^k\| &\leq \int_0^1 \|Df(x^k)^{-1}(Df(x^{k-1} + t(x^k - x^{k-1}))) - Df(x^{k-1})\| \|x^k - x^{k-1}\| dt \\ &\leq \int_0^1 \gamma_k t \|x^k - x^{k-1}\|^2 dt = \frac{\gamma_k}{2} \|x^k - x^{k-1}\|^2 \\ &\leq \frac{\gamma_k}{2} \left(\frac{h_{k-1}}{\gamma_{k-1}}\right)^2 = \frac{h_k}{\gamma_k} \leq \frac{1}{2^k} \alpha.\end{aligned}$$

(Third line: (d) for $k-1$, then the definition of h_k ; last $\leq$: Lemma (iii).)

So (a)–(d) hold for k , and by induction for all k : Part 1 and the step bound $\|x^{k+1} - x^k\| \leq h_k/\gamma_k$ are proved.

To be continued (next lecture): Cauchy $\Rightarrow x^* = \lim x^k$ exists in $\overline{S_r(x^0)}$; $f(x^*) = 0$; quadratic rate; Part 3.

Details: Reading the Assumptions

Each hypothesis is a measurable quantity at the *starting point*:

- $\alpha \geq \|Df(x^0)^{-1}f(x^0)\| = \|x^1 - x^0\|$: the length of the first Newton step.
Small α : we start near a solution, as measured by Newton itself.
- γ : the affine-invariant Lipschitz constant of Df , a measure of curvature (nonlinearity). In the limit $\gamma \rightarrow 0$, Df is constant, f is affine, and Newton finishes in one step.
- $h = \alpha\gamma < \frac{1}{2}$: “first step $\times$ nonlinearity” is small: the tangent model is trustworthy over the region the iterates explore.

The radius satisfies $\alpha \leq r \leq 2\alpha$. For $0 \leq h < \frac{1}{2}$,

$$\sqrt{1-2h} \leq 1-h \quad (\Leftrightarrow 1-2h \leq 1-2h+h^2),$$

$$1-2h \leq \sqrt{1-2h} \quad (\text{as } 0 \leq 1-2h \leq 1),$$

so $h \leq 1 - \sqrt{1-2h} \leq 2h$ and $\alpha = \frac{h}{\gamma} \leq r \leq \frac{2h}{\gamma} = 2\alpha$: the whole iteration happens within about two first steps of x^0 .

Details: Why the Step Bound Is Quadratic

In the proof, $x = x^{k-1}$, $y = x^k$, $\Delta = x^k - x^{k-1}$, and the Newton equation says $Df(x^{k-1})\Delta = -f(x^{k-1})$. Subtracting it from the identity leaves a *difference of Jacobians*:

$$x^{k+1} - x^k = -Df(x^k)^{-1}f(x^k) = -\int_0^1 Df(x^k)^{-1}(Df(x^{k-1} + t\Delta) - Df(x^{k-1}))\Delta dt.$$

This is exactly what the Lipschitz condition (c) can bound: the integrand has norm at most $\gamma_k t \|\Delta\| \cdot \|\Delta\|$. Then

$$\int_0^1 t dt = \frac{1}{2} \implies \|x^{k+1} - x^k\| \leq \frac{\gamma_k}{2} \|\Delta\|^2.$$

The new step is bounded by the *square* of the old one. This one computation is the heart of the whole proof.

The Scalar Majorant Sequences at a Glance

$$h_k = \frac{1}{2} \frac{h_{k-1}^2}{(1 - h_{k-1})^2}, \quad \gamma_k = \frac{\gamma_{k-1}}{1 - h_{k-1}}, \quad h_0 = h, \quad \gamma_0 = \gamma.$$

- $h < \frac{1}{2} \Rightarrow h_k < 2h_{k-1}^2 < h_{k-1}$: the h_k decrease. In fact $2h_k < (2h_{k-1})^2$, so $2h_k < (2h)^{2^k}$: doubly exponential decay (they square each time).
- γ_k increases but stays bounded: $\gamma_k \nearrow \gamma / \sqrt{1 - 2h}$, because $\sqrt{1 - 2h_k} / \gamma_k$ does not change with k .
- The step bound $\|x^{k+1} - x^k\| \leq h_k / \gamma_k \leq \alpha / 2^k$ makes the iterates a Cauchy sequence (next lecture), and the sum $\sum_{j < k} h_j / \gamma_j = 1 / \gamma - 1 / \gamma_k < r$ keeps every iterate inside the ball.

Big picture: the vector iteration is dominated (“majorized”) by a scalar recursion that we can analyze completely. This proof pattern – majorant sequences – recurs throughout numerical analysis.

Looking Ahead

- **Next lecture:** Newton's Method in Practice; Damping